Rental Insights API

A RESTful API for Rental Listings and Market Insights

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Course: COMP3011 Coursework 1

Date: March 2026

Project Overview

Core Functionality

Implements CRUD operations for rental listings and geographic areas, providing a solid foundation for data management.

Market Insights

Provides analytics endpoints to calculate rent distribution and affordability indicators, enabling data-driven decisions.

Official Data Integration

Incorporates VOA Private Rental Market Statistics (2018-2019) for authoritative comparisons and benchmarking.

Key Deliverables

Complete package including a public GitHub repo, Swagger API documentation, a detailed technical report, and this presentation.

Architecture & Module Structure

Core Tech Stack

FastAPI Framework
High performance with automatic OpenAPI documentation.

SQLAlchemy & SQLite
Lightweight ORM with file-based storage for easy demonstration.

Project Module Layout

main.py- Entry point & Router registration

models.py- DB schema definitions (Areas, Listings)

schemas.py- Pydantic models for data validation

routers/- Modular API endpoints (listings, analytics)

scripts/- Utilities for data import & checks

Data Model

Areas Table

- Stores geographic data (name, postcode, city)
- Includes average income for affordability calculations

Listings Table

- Contains user-submitted rental ads (rent, bedrooms, type)
- Linked to Areas via**area_id**foreign key

OfficialRentStats

- Imported VOA PRMS data (median, mean, quartiles)
- Provides statistical benchmarks for analysis

Relationship: Establishes a One-to-Many relationship between Areas and Listings.

Core Features & Demo

CRUD Operations

Full RESTful endpoints for managing listings and areas: Create, Read, Update, Delete.

Analytics Insights

- Area rent stats (mean/median)
- Rent-to-income ratio analysis

Official Stats

- Query imported VOA data
- Compare user data with official benchmarks

Demo: Access interactive documentation at**http://127.0.0.1:8000/docs**

Technology Choices & Justification

FastAPI

Chosen for automatic request validation (Pydantic), high performance (ASGI), and auto-generated Swagger docs, reducing development time significantly.

SQLAlchemy + SQLite

SQLAlchemy provides a clean ORM layer. SQLite is ideal for coursework as it's file-based, requires no server setup, and is easily portable.

pandas + xlrd

Used to parse the legacy .xls format of the VOA dataset and import it into the relational database, enabling efficient querying.

Challenges & Future Work

Challenges & Solutions

Importing .xls Data
Parsed legacy format with pandas/xlrd and handled missing values.

Python Import Paths
Fixed 'module not found' errors by configuring sys.path.

Case Sensitivity
Implemented case-insensitive queries for better UX.

Limitations & Future Improvements

Performance Optimization
Move analytics calculations to the database for larger datasets.

Security Enhancement
Add API key/JWT authentication and rate limiting.

Data Freshness
Automate the download of the latest VOA datasets.

Thank you for watching!

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